AI Engineering

Date:

Assignment # 01

$d_k = 2$, $d_v = 2$

$$Q, K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$S = QK^T/\sqrt{d_k}$$

$$\Rightarrow \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 0.7071 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}$$

$$\text{Softmax} = \frac{e^x}{\Sigma e^x}$$

$$\Rightarrow \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 3 & 4 \\ 3.4066 & 4.4066 \\ 3.5107 & 4.5108 \end{bmatrix}$$

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